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TEMARL

Transformer-Enhanced Multi-Agent Reinforcement Learning for Adaptive Honeypot Deception

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The system works. The architecture claim does not.

✓ What we proved

The deception system contains attackers **60.4%** of the time, versus **54.8%** for a defender with no attacker model ($p = 0.0004$).

The Transformer performs **statistically identically** to GRU and LSTM — using **44%** and **35%** of their parameters.

✗ What we could not prove

That the Transformer **beats** GRU or LSTM at reading attacker behaviour.

Tested 8 pre-registered ways, plus a powered replication at $n = 41$. Every result null. We report this honestly.

Four figures that summarise the thesis

0.6040

System containment rate
(vs 0.5475 with no attacker model)

62–77%

of the maximum achievable
gain, captured

44.4%

Transformer parameters
relative to GRU

p=0.0001

Statistical equivalence
proven, canonical map

Dataset

CAM-LDS: 36 runs, 1,347 attacker actions, 7 campaigns, 82 techniques.

Effective sample size: 3.09.

Environments

Three built: frozen control, variable-topology, and a repaired version after we found the original could not test the hypothesis.

Models

Six encoders compared under identical conditions: Transformer, GRU, LSTM, GRU-d32, SetEncoder, plus two network encoders.

Our first experiment could not have answered its own question

We built a **reference ladder** to check whether the task could reward sequence modelling at all. The key measure:

Sequence value = (perfect next-step prediction) – (perfect campaign label).
If this is zero, **no** sequence model can distinguish itself — however good it is.

Environment	Sequence value	Models beating a no-model baseline	Episode length
Original (experiment v4)	+0.0005	1 of 6	5.5 steps
Repaired (experiment v5)	+0.0800	5 of 5	18.1 steps

Cause: the campaigns had been **clustered by their own correct answer**, and the transition matrix had a bug that discarded 87% of the sequence information. Only the second row is a fair test.

What we did to the data

Rebuilt from source, audited twice

- Re-parsed all 36 published playbooks instead of using a hand reconstruction.
- Wrote a **second independent parser** to check the first — it found 3 more bugs.
- **5 defects fixed**; 8 lost actions recovered (1,339 → 1,347).
- Vocabulary extended 78 → 84: **6.20%** of actions had been silently discarded as unknown.

Three of our own claims withdrawn

Earlier claim	Truth
Counts match the paper "exactly"	82 vs 81 — the match was a bug artefact
Reconstruction "invented" T1095	T1095 is real; precision is 100%
COMISET is 1156× bigger	≈18× — 98.5% are repeats

Clustering: 7 campaigns were grouped into 3 classes *by which decoy was best for them*. That made knowing the group equal to knowing the answer — the root cause of the failure.

MAIN RESULT

Six architectures, one outcome

Repaired environment · 10 paired seeds · 200 test episodes · all models passed the validity gate

History model	Network model	Canonical	Trained sizes	Unseen structure	Unseen sizes	Parameters
Transformer-RoPE	DeepSets	0.6540	0.6040	0.6025	0.4865	156,310
GRU	DeepSets	0.6550	0.6100	0.6080	0.4820	351,702
LSTM	DeepSets	0.6560	0.6085	0.6115	0.4845	450,518
GRU-d32 (control)	DeepSets	0.6520	0.6055	0.6050	0.4920	236,598
Transformer-RoPE	EntityTransformer	0.6525	0.6075	0.6070	0.4810	156,310
GRU	EntityTransformer	0.6565	0.6175	0.6135	0.5000	351,702
<i>Reference: no attacker model</i>		0.5920	0.5475	0.5470	0.4465	—

Six architectures span **0.0045** on the canonical map. All beat the no-model baseline by +0.057 to +0.070 ($p \leq 0.0018$).

From "no difference found" to "equivalence proven"

Why this matters

A non-significant test cannot separate "*there is no effect*" from "*our study was too small*".

A **TOST equivalence test** can. It makes a positive claim: the difference is **demonstrably below a stated threshold**.

Transformer $\equiv$ GRU in 3 of 4 conditions

Transformer $\equiv$ LSTM in **all 4** conditions
at **44.4%** and **34.7%** of their parameters

Equivalence results (margin 0.02)

Condition	vs GRU	vs LSTM
Canonical map	0.0001 ✓	0.0029 ✓
Trained sizes	0.0124 ✓	0.0118 ✓
Unseen structure	0.0029 ✓	0.0432 ✓
Unseen sizes	0.0527	0.0304 ✓

Values are p-values for equivalence; ✓ means equivalence is statistically established.

We gave the one promising lead a proper second chance

One of the eight tests nearly reached significance. Rather than claim it, we calculated the required sample size, **pre-registered a replication at $n = 31$** , ran it once on fresh seeds, and reported the outcome.

Run	n	Entity attention	Baseline	Difference	Effect size	p
Original finding	10	0.5000	0.4820	+0.0180	+0.61	—
Pre-registered replication	31	0.4597	0.4652	−0.0055	−0.17	0.826
Pooled final estimate	41	0.4695	0.4693	+0.0002	+0.01	0.481

Four times a promising small-sample result vanished when tested at proper sample size. Seed-to-seed variation ($SD \approx 0.068$) was larger than every effect ever claimed from small samples. **This is itself a reportable finding.**

Why the architecture claim failed — five reasons

#	Reason	Evidence
1	The dataset is tiny. Transformers need large data; this has 7 campaigns.	Effective n = 3.09
2	The sequences are short. Attention's advantage comes from long-range dependencies that do not exist at 18 steps.	18.4 steps
3	Clustering removed the sequence problem , turning it into 3-way classification.	+0.0001
4	An estimator bug discarded 87% of the sequential structure in the data.	0.627 of 4.83 bits
5	The task is nearly saturated. Both models already capture 62–77% of the maximum, leaving less headroom than random noise.	~0.02 remaining

Important: reasons 1, 2 and 5 are properties of the *problem*, not defects in the Transformer. The model is not broken — the task does not contain enough sequential structure for attention to demonstrate an advantage.

What this thesis contributes

1 · A working system

First Transformer-enhanced MARL honeypot deception system grounded in MITRE D3FEND with CAM-LDS attacker profiles. Captures **62–77%** of the maximum achievable gain. Attacker-intent modelling is worth **+0.148 to +0.171** containment.

2 · Proven parameter efficiency

Statistically demonstrated equivalence to GRU and LSTM at **44.4%** and **34.7%** of their parameters. Not a null result — a positive equivalence claim, $p = 0.0001$.

3 · A methodological finding

A pre-registered environment gate **can pass while the environment is useless** for the hypothesis being tested. Architecture comparisons need a **reference ladder** and a **validity gate** — without them, v4 produced 8 null results from 5 models that had never learned.

4 · Data contributions

A source-verified CAM-LDS grounding with **5 parser defects** found and fixed, verified independently; a quantified estimator defect that destroyed **87%** of the corpus's sequential information.

Every hypothesis was registered before the experiment ran

Commit	Content	Timing
7499d29	v4 pre-registration	Before the v4 run
1413afb	v5 pre-registration + environment repair	Before the v5 run
0c25e12	v5 result — all 8 contrasts reported	After
7261a4a	Equivalence tests + v6 pre-registration	Before the v6 run
4b53a26	v6 replication result	After

What we did

- Reported every declared test, including those that failed
- Recorded two of our own hypotheses that proved false
- Withdrew four claims when better measurement contradicted them

What we refused to do

- Add seeds until a result became significant
- Drop a test that failed
- Re-test v5 one-sided to turn $p = 0.690$ into $p = 0.043$

The honest version of this thesis is the defensible one

The title's implied claim — that attention is what makes this system work — is not supported. What *is* supported is more precise:

A Transformer-based honeypot deception system that captures 62–77% of the achievable defensive gain, and matches recurrent alternatives using roughly one-third of their parameters — established under pre-registration, on an instrument we first had to prove was capable of measuring it.

Limitation

Trained by imitation learning, not reinforcement learning — under PPO no model beat the baseline, and why remains open.

Limitation

One environment, one scripted attacker. A learning attacker could change the conclusions.

Future work

Test entity attention on genuinely varying topologies with more power; resolve the PPO failure.